

Due: March 11th, before class.

Name: Solution

Collaborators:

Document your work and note any collaborators. In your write-up, please neatly document any code you have written, alongside its output, and including any necessary commentary to answer the question. Please use an `.ipynb` notebook for any code, include commentary in markdown code blocks, and print the `.ipynb` notebook to PDF to attach to your submission. The L^AT_EXsource for this file may be helpful: <https://www.overleaf.com/read/rqqzmckbtcbc#6e86f6>.

Problems:

Theorem 1. Suppose that $f(x)$ has a continuous second derivative in $[a, b]$ (often expressed as $f \in C^2[a, b]$). Let $p_n(x)$ be the piecewise linear interpolant of $(t_i, f(t_i))$ for $i = 0, \dots, n$ where $t_i = a + ih$ and $h = (b - a)/n$. Then

$$\|f - p_n\|_\infty = \max_{x \in [a, b]} |f(x) - p_n(x)| \leq Mh^2,$$

where $M = \|f''\|_\infty$.

1. We can prove the theorem above using the mean value theorems from calculus. If f is continuously differentiable in (a, b) , then there exists points s and t in (a, b) such that

$$\int_a^b f(z)dz = (b - a)f(s) \quad \text{and} \quad f'(t) = \frac{f(b) - f(a)}{b - a}.$$

For the problems below, assume $x \in (t_k, t_{k+1})$.

- (a) Show that for some $s \in (t_k, t_{k+1})$,

$$f(x) = f(t_k) + (x - t_k)f'(s).$$

Assuming that f is continuously differentiable in (t_k, t_{k+1}) , then it is continuously differentiable in (t_k, x) . Applying MVT on the interval yields that $\exists s$ such that $f'(s) = \frac{f(t_k) - f(x)}{t_k - x}$. Rearranging yields that desired result.

- (b) Show that for some other values u and v in (t_k, t_{k+1}) ,

$$f'(s) - \frac{f(t_{k+1}) - f(t_k)}{t_{k+1} - t_k} = (s - u)f''(v).$$

By last question we know that if we let $x = t_{k+1} \exists u$ such that

$$f'(u) = \frac{f(t_{k+1}) - f(t_k)}{t_{k+1} - t_k}.$$

Assuming $f \in C^2[t_k, t_{k+1}]$, we can apply MVT again on f' to yield that $\exists s$ such that

$$f''(v) = \frac{f'(s) - f'(u)}{s - u}$$

for $v \in (u, s) \subseteq (t_k, t_{k+1})$ ¹. Plugging in $f'(u)$ from above and rearranging produces the desired result.

(c) Use

$$p_n(x) = f(t_k) + \frac{f(t_{k+1}) - f(t_k)}{t_{k+1} - t_k}(x - t_k) \quad \text{for } x \in [t_k, t_{k+1}]$$

to finish the proof of the theorem above.

We know that

$$f(x) - p_n(x) = f(x) - f(t_k) - \frac{f(t_{k+1}) - f(t_k)}{t_{k+1} - t_k}(x - t_k).$$

Now by (a) $f(x) = f(t_k) + (x - t_k)f'(s)$ for some s , so

$$\begin{aligned} f(x) - p_n(x) &= (x - t_k)f'(s) - \frac{f(t_{k+1}) - f(t_k)}{t_{k+1} - t_k}(x - t_k) \\ &= (x - t_k) \left(f'(s) - \frac{f(t_{k+1}) - f(t_k)}{t_{k+1} - t_k} \right). \end{aligned}$$

Now by (b) the quantity in the brackets is $(s - u)f''(v)$ for $u, v \in (t_k, t_{k+1})$. Thus

$$f(x) - p_n(x) = (x - t_k)(s - u)f''(v).$$

Taking the infinity norm on both sides yield

$$\|f(x) - p_n(x)\|_\infty \leq \|x - t_k\|_\infty \|s - u\|_\infty \|f''(v)\|_\infty.$$

This because the max of a product is less than or equal to the product of the max of the individual components (the product could only reach the product of the max if and only if the components all peak at one place).

Looking at each piece, we know that $x - t_k$ and $s - u$ are both bounded above by h , the size of the interval, since all the individual quantities are in the interval. $\|f''(v)\|_\infty \leq M$ by definition. Thus we can see that

$$\|f(x) - p_n(x)\|_\infty \leq h^2 M,$$

as desired.

¹It is not necessarily true that $s > u$ but if it is the other way round the equation is still right.

2. (a) Show using L'Hôpital's Rule that the centered formula approximation

$$f'(0) \approx \frac{f(h) - f(-h)}{2h}$$

converges to an equality as $h \rightarrow 0$.

By L'Hôpital's rule, since we have a $0/0$ situation,

$$\lim_{h \rightarrow 0} \frac{f(h) - f(-h)}{2h} = \lim_{h \rightarrow 0} \frac{f'(h) + f'(-h)}{2} = \frac{2f'(0)}{2} = f'(0).$$

In the first equality we used chain rule.

- (b) Now, consider the finite-difference formula

$$f'(x) \approx \frac{1}{h} \sum_{k=-p}^q a_k f(x + kh)$$

where $h \rightarrow 0$. Suppose we require convergence of the formula to $f'(x)$ in the limit $h \rightarrow 0$. What does this require of the finite difference weights $\{a_k\}$? (Hint: Considering what is required in order to apply L'Hôpital's Rule, as well as the result of applying it.)

We want

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \sum_{k=-p}^q a_k f(x + kh)$$

to be true. In order to have a well-defined limit, we must have

$$\lim_{h \rightarrow 0} \sum_{k=-p}^q a_k f(x + kh) = \sum_{k=-p}^q a_k f(x) = 0.$$

Assuming arbitrary $f(x) \neq 0$, then we must have $\sum_{k=-p}^q a_k = 0$. Now we can apply

L'Hôpital's rule to get

$$\lim_{h \rightarrow 0} \frac{1}{h} \sum_{k=-p}^q a_k f(x + kh) = \lim_{h \rightarrow 0} \frac{\sum_{k=-p}^q k a_k f'(x + kh)}{1} = \sum_{k=-p}^q k a_k f'(x).$$

If this is equal to $f'(x)$ for arbitrary $f'(x) \neq 0$, then we must have $\sum_{k=-p}^q k a_k = 1$.

Lab: We derive finite difference formulas by interpolating with a polynomial, then differenti-

ating the interpolant. This leads to, for example, the formulas

$$W_1(h) = \frac{f(h) - f(0)}{h} = f'(0) + O(h), \quad (1)$$

$$W_2(h) = \frac{f(h) - f(-h)}{2h} = f'(0) + O(h^2), \quad (2)$$

$$W_4(h) = \frac{-f(2h) + 8f(h) - 8f(-h) + f(-2h)}{12h} = f'(0) + O(h^4). \quad (3)$$

The terms $O(h^p)$ are a statement about the accuracy of the formula, which we say has order p . However, interpolation by a polynomial isn't ideal for every f you might encounter. One reason is that polynomials are smooth—they have infinitely many derivatives. Consider $f(x) = |x|$, for example. It's continuous but not differentiable at $x = 0$. What value should the finite difference attempt to get there: $+1$, -1 , or something else? For a function that has one derivative but not two, there is a unique value to converge to, but the convergence rate is slowed by the singularity.

Goals: You will apply finite difference methods of different orders to functions that lack smoothness at a point, and observe the effects on the order of accuracy.

Preparation: Determine the number of continuous derivatives at $x = 0$ for the following functions:

$$g_1(x) = \begin{cases} 5x + 1, & x \leq 0 \\ (x + 1)^5, & x > 0 \end{cases} \quad (4)$$

$$g_2(x) = \begin{cases} 10x^2 + 5 \sin(x) + 1, & x \leq 0 \\ (x + 1)^5, & x > 0 \end{cases} \quad (5)$$

Procedure: Perform the following steps.

- (a) Define $f(x) = \exp(\sin(x + 1))$. This is our “control” case of a function with infinitely many derivatives everywhere. For each value of $h = 2^{-1}, 2^{-2}, 2^{-3}, \dots, 2^{-10}$, compute $W_1(h)$, $W_2(h)$, and $W_4(h)$, storing the results in vectors. Then make a log-log plot showing all three cases of $|W_i(h) - f'(0)|$ as functions of h . You should see three straight lines with different slopes.
- (b) To the graph from step 1, add plots of the functions h^1 , h^2 , h^3 , and h^4 , as dashed lines. (Use labels, legends, etc.) Confirm that each solid convergence curve matches well with the dashed line corresponding to the order of accuracy in the presentation of the formulas above.
- (c) Now input g_1 from (4), by defining

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g1 = x -> (5*x+1).*(x<=0) + ((x+1).^5).*(x>0);
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(This defines the function piecewise for negative and positive x .) Make a plot of the function over $[-1/4, 1/4]$.

- (d) Repeat steps (a)–(b) for the function in step (c). Now what are the observed orders of accuracy for W_1 , W_2 , and W_4 ?
- (e) Repeat steps (a)–(b) for the function g_2 in (5).
- (f) What number of continuous derivatives does g_1 and g_2 have?
- (g) Hypothesize how the number of continuous derivatives of the input function affects the order of accuracy of finite difference methods. Comment on how your hypothesis is corroborated by the three convergence plots created above.